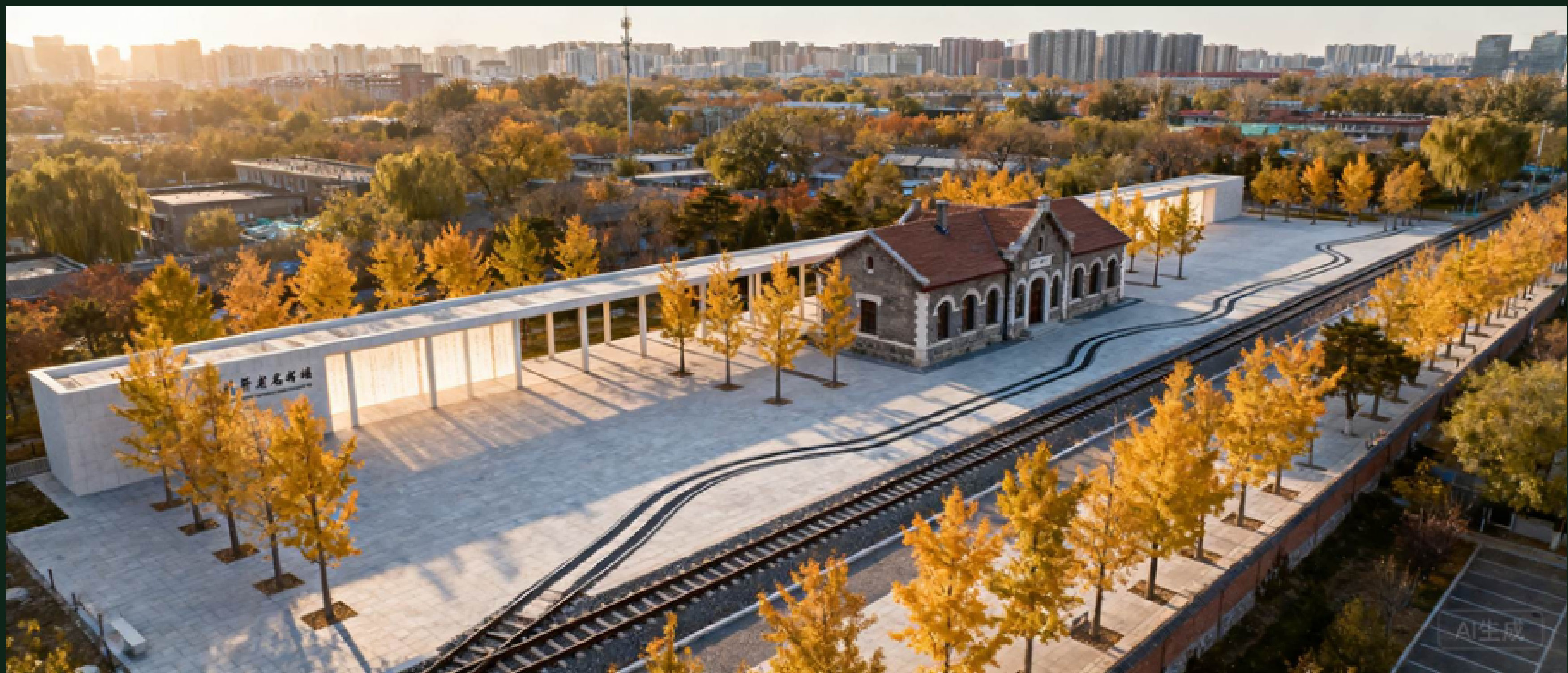


THE EMERGENT BELT

涌现之带

Jing-Zhang Hyper Line

Centennial Jing-Zhang AI Belt · open-call submission

KUN-SAL Spatial Quant Lab · Xing Xinhua

λc=0.42 connection	H=1.556 health
D=1.746 fractal	25,476 POIs
+15.9% job reach	44亿 investment

Conceptual: provisional boundary dashed only; statutory indicators pending

01

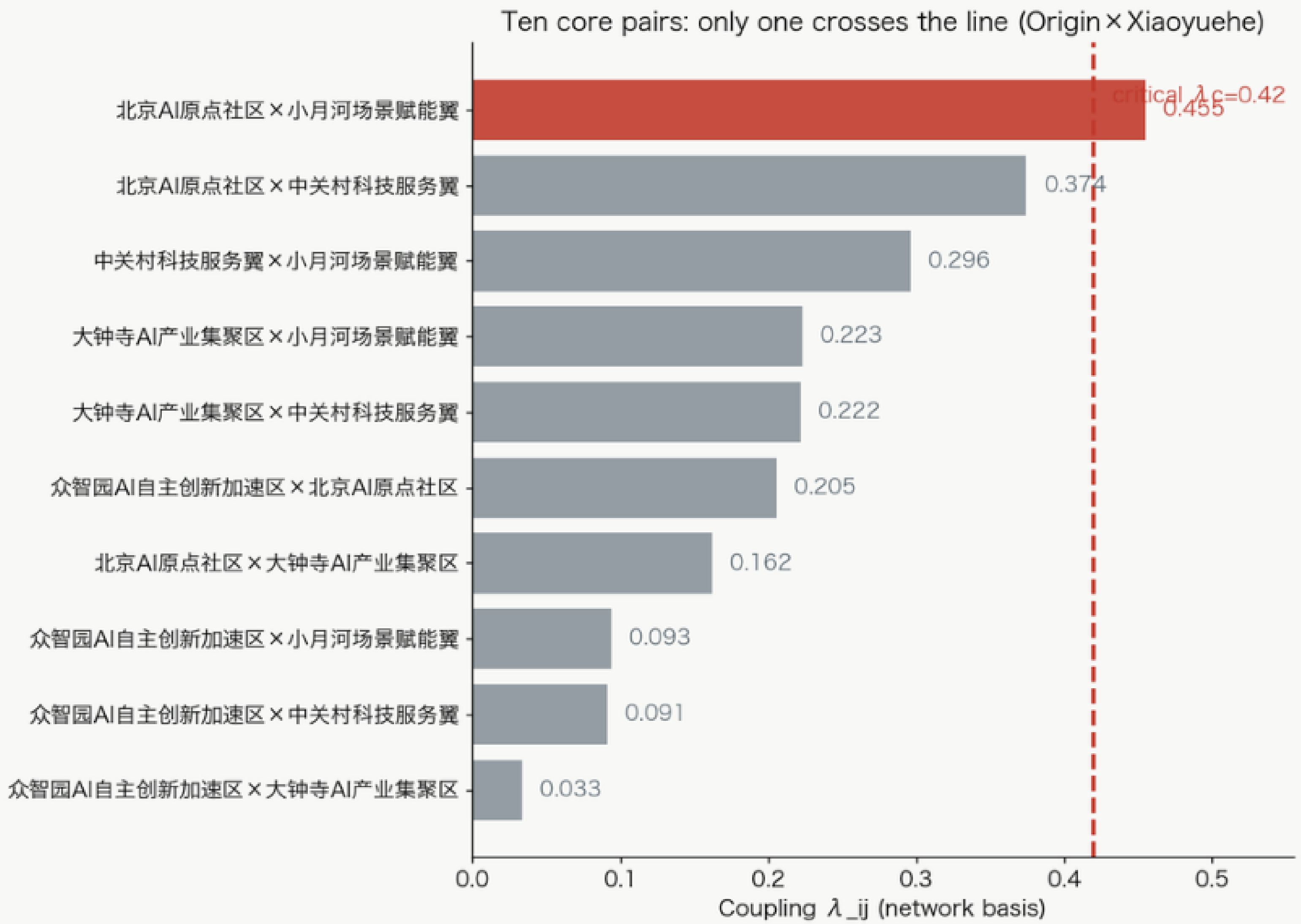
Diagnosis: what blocks this belt

Five scientists' consultation · Space syntax · LUTI (jobs-housing dynamics) · POI dynamics · Boundary correction

① Spatial interaction λ: stuck subcritical

Fig C1 · Wilson spatial-interaction λ calibration (status quo)

$\lambda\_ij = \exp(-d\_ij/2500m)$; $\lambda > \lambda c=0.42$ = reachable within the 10-min innovation circle (OSM network)



Supercritical share · euclidean	Supercritical share · network	5-core global connectivity
50.00%	29.17%	No (severed)

Diagnosis: the five cores are not connected under the 10-min coupling — a healthy skeleton severed by seven roads (λ is cut one of the diagnosis ring).

- Only one of ten core pairs crosses the line (Origin × Xiaoyuehe 0.455)
- Zhongzhiyuan×Dazhongsi λ=0.033: adjacent on the map, unreachable in ten minutes
- Not without vitality — vitality chopped apart by seven roads
- The leverage point is the seven gaps; no even spreading

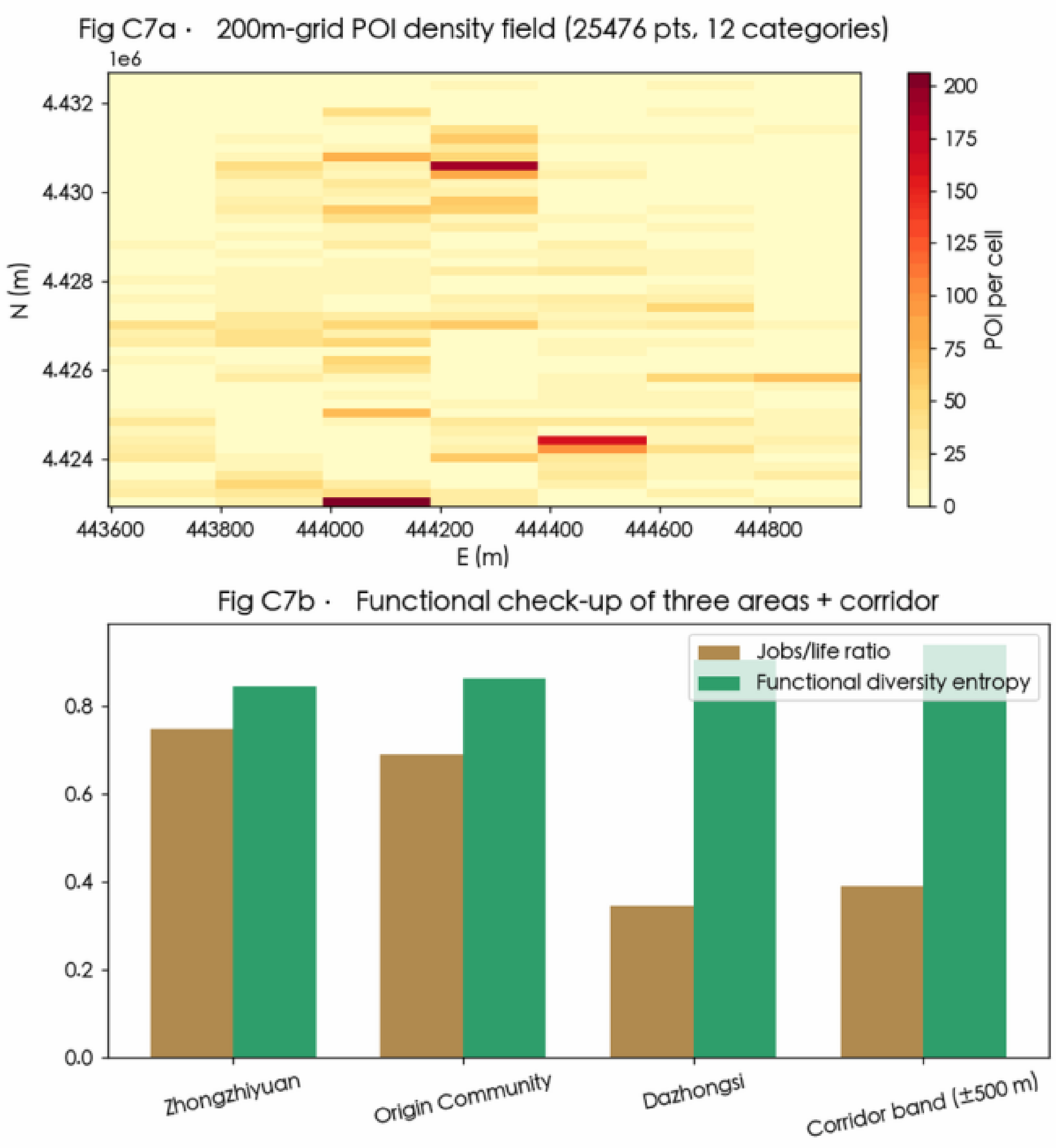
Basis: λ=exp(-d/2500m) proxy; recalculation on official data

29.17%	2/5	0.42
supercritical share	cores connected	critical λ

② POI check-up: demand recognized by 25,476 points

- 25,476 POIs across the belt (2026-08-15 snapshot)
- Life services over-supplied; innovation content under-supplied
- Finance −63.0% / research −45.9% / Origin medical 2.88/km², the lowest
- Dazhongsi 511/km², the densest; jobs/life 0.344: life dense, work thin

Fig C7 · Amap POI dynamics: the invisible engine of the city



Data: Amap aggregate indicators; no raw points published

25,476	−63.0%	−45.9%
POIs	finance gap	research gap

③ Space syntax: configuration shapes movement

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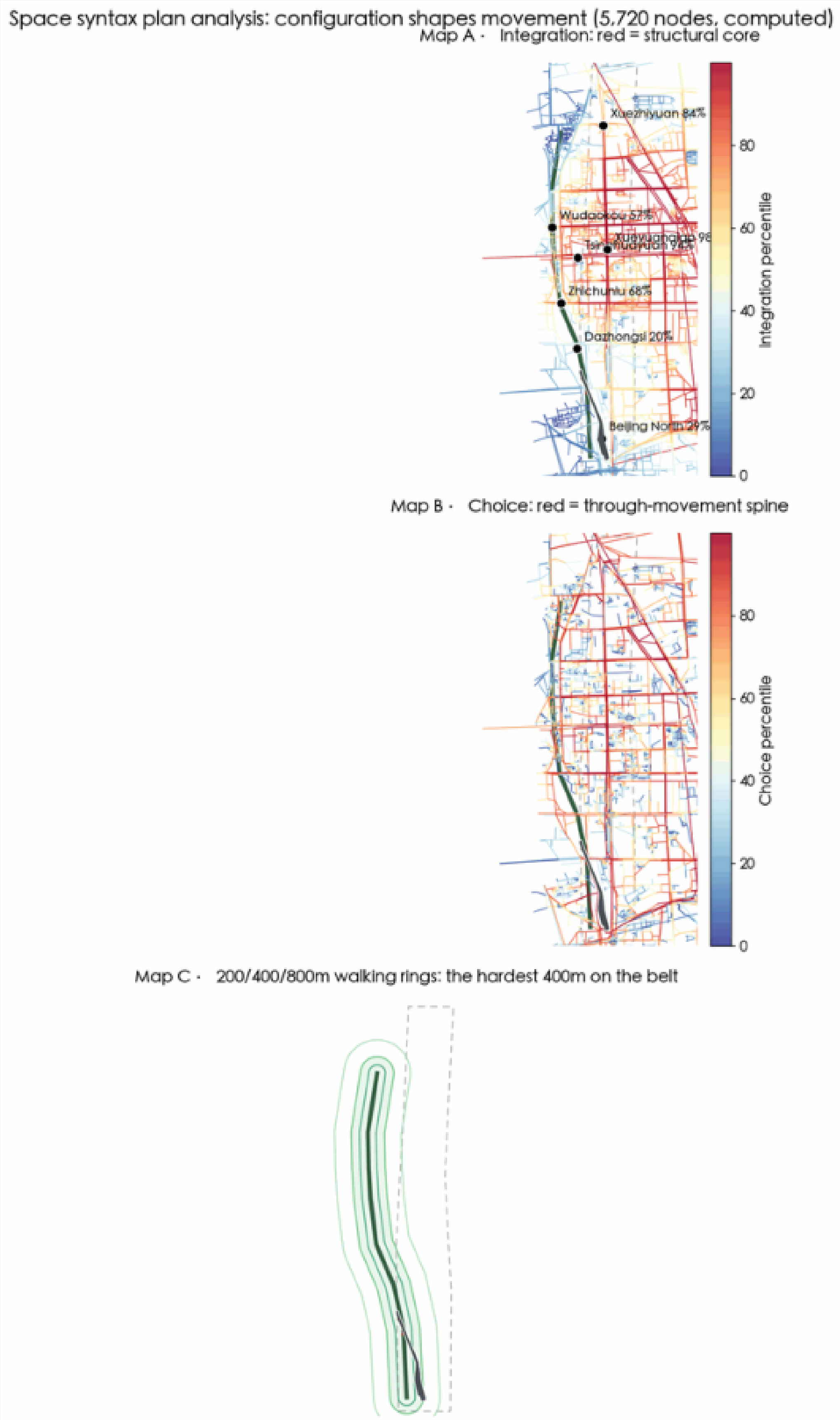
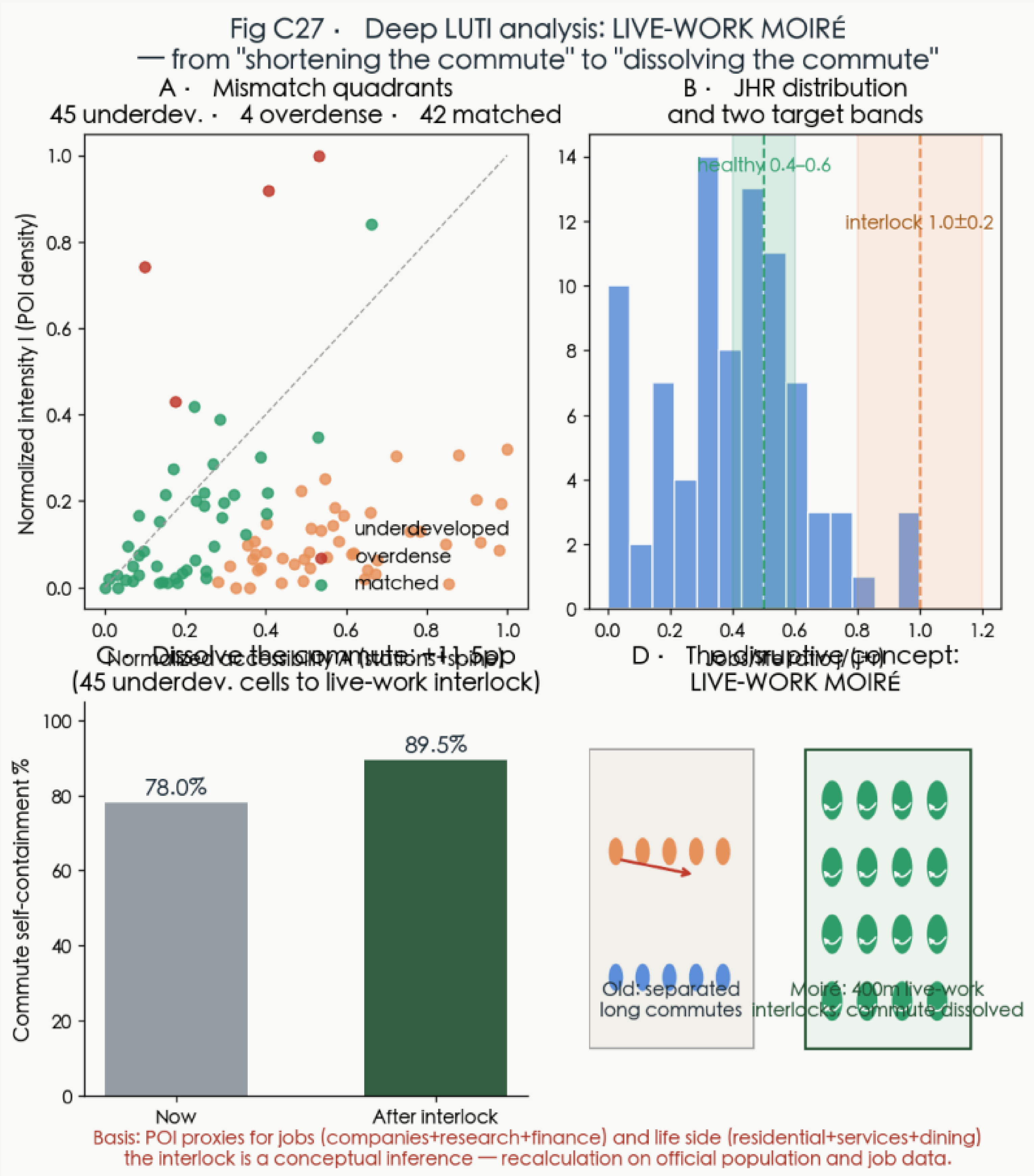


- Xueyuanqiao 97.5%: the belt's mouth at the N4th Ring × Xueyuanlu
- Tsinghuayuan 93.5%: a 1909 siting still the structural core
- Dazhongsi 20.2%: syntactic periphery, vitality must come from function
- After stitching: Dazhongsi +70%, Beijing North +49%, Wudaokou +43%

5,720-node network, computed; low intelligibility is normal for organic fabric

93.5%	20.2%	+70%
Tsinghuayuan int.	Dazhongsi int.	stitching gain

⑤ Deep LUTI: LIVE-WORK MOIRÉ — dissolve the commute



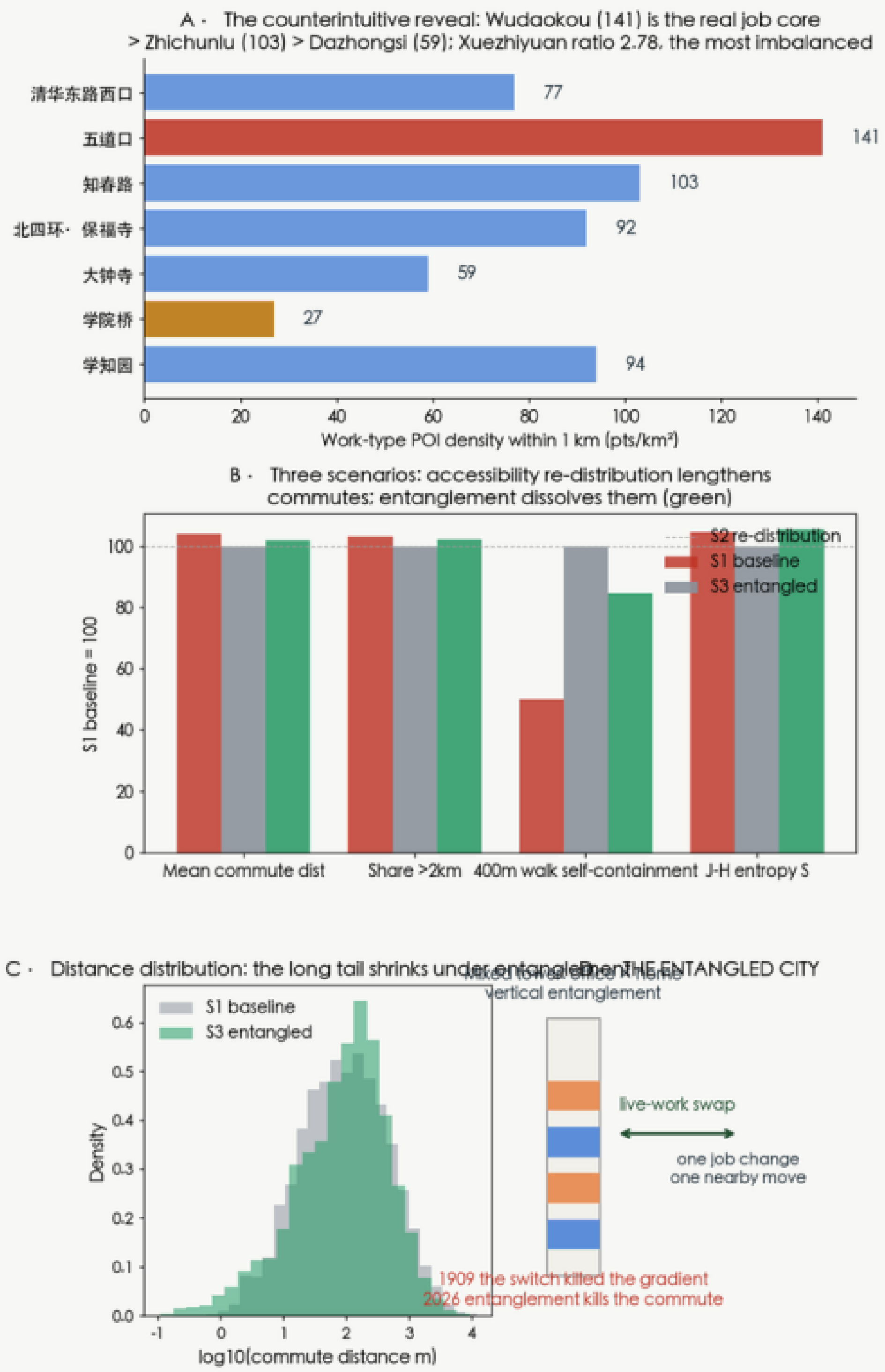
- Disruptive concept: 45 underdeveloped cells rebuilt as 1.0±0.2 live-work interlocks
- Commute self-containment 78.0% → 89.5% (+11.5pp, computed)
- Mismatch quadrants 45/4/42: where it's wrong → after stitching → what remains
- Not a city with fewer commutes — a city that does not need commuting

POI proxies: recalculation on official population and job data

+11.5pp commute dissolved	78→89.5% self-containment	1.0±0.2 interlock ratio

⑥ Three-layer LUTI scenarios: concentration lengthens, entanglement dissolves

Fig C28 · Three-layer LUTI scenario runs: only live-work entanglement dissolves the commute (Batty-style scenarios)



- The reveal: Wudaokou 141/km² is the real job core; Xuezhiyuan ratio 2.78, most imbalanced
- Three scenarios (gravity OD): concentration +3.9% distance, walking self-containment 7.0%→3.5%
- Entangled: entropy S +5.3% (more mixed); rules = paired infill + mixed towers + S in the annual check-up
- 1909 the switch killed the gradient, 2026 entanglement kills the commuting tide

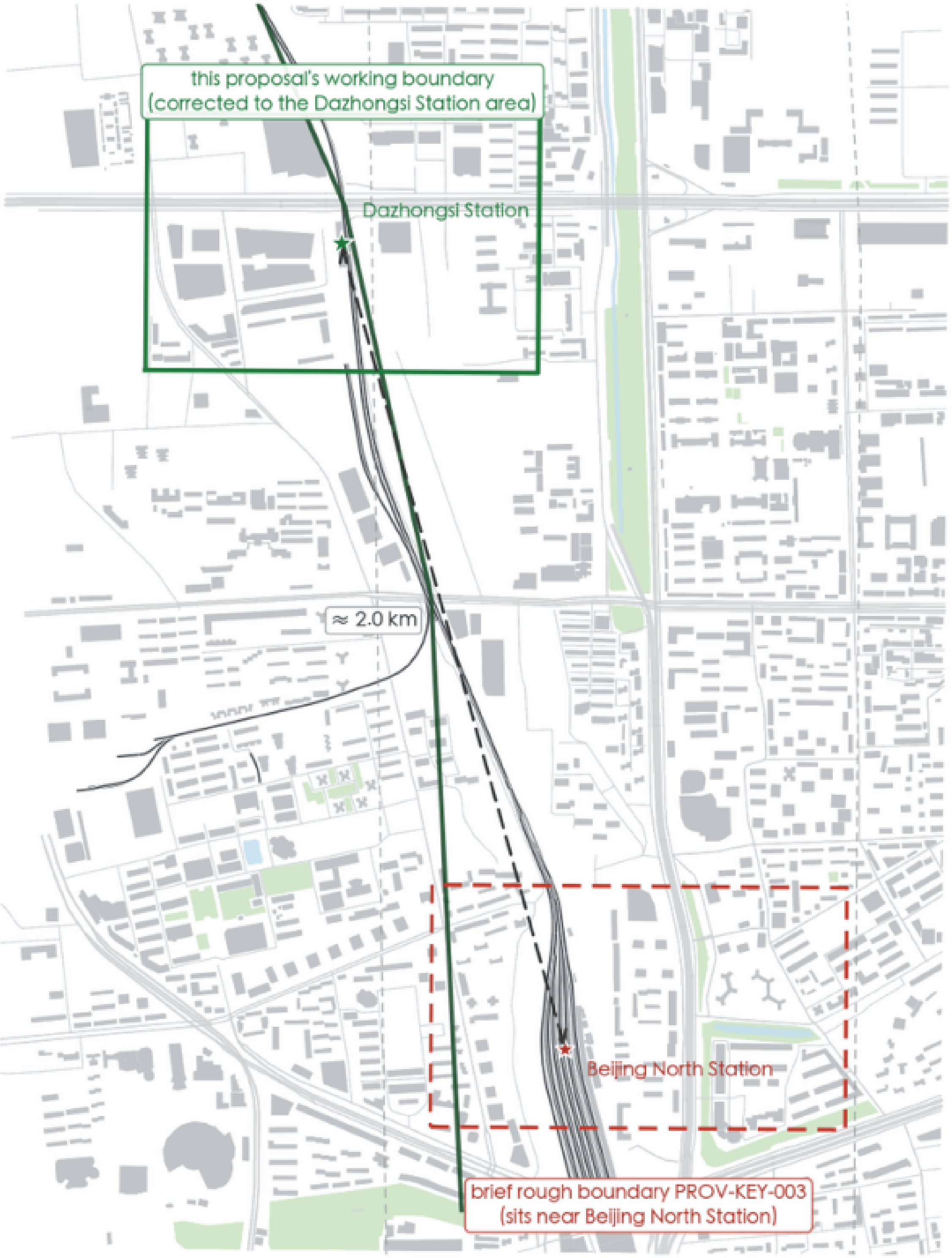
Gravity OD + POI proxies; recalculation on official OD; floorspace ledger = the brain's first new database

Basis: production-constrained gravity OD (β=1/2500m, same basis as λ); POI proxies for jobs/housing; S3 is a conceptual scenario — recalculation on official commute OD (signalling/card data).

141/km²	-3.5pp	+5.3%
Wudaokou job core	concentration cost	entropy gain

⑦ Boundary correction: fixing the brief's rough boundary

Fig C18 · Evidence for A-BOUNDARY-003



Evidence ① · the brief's own words

provisional_boundaries_basis.md L26:
"Dazhongsi AI Industry Cluster | the third southern area; the tasks involve Dazhongsi Station | 72.0 ha"

Evidence ② · data vs name

yet the supplied polygon actually sits near Beijing North (39.944–39.950), ~2 km from Dazhongsi Station (39.9653)

Evidence ③ · the gap is admitted

missing-data.md L8: the precise polygons of the three key areas are missing

Response: correct the working boundary by name semantics and recompute everything;
disclose > quietly fix (A-BOUNDARY-003)

- The brief's PROV-KEY-003 rough boundary sits near Beijing North (39.944–39.950)
- Its own basis table names Dazhongsi Station (39.9653) — ~2 km away
- missing-data.md: precise polygons are missing
- Corrected by name semantics, assumption A-BOUNDARY-003, full recalculation on official data

Integrity first: a data mismatch must be disclosed, never quietly fixed

≈2 km distance	72.3 ha working boundary	A-003 assumption id

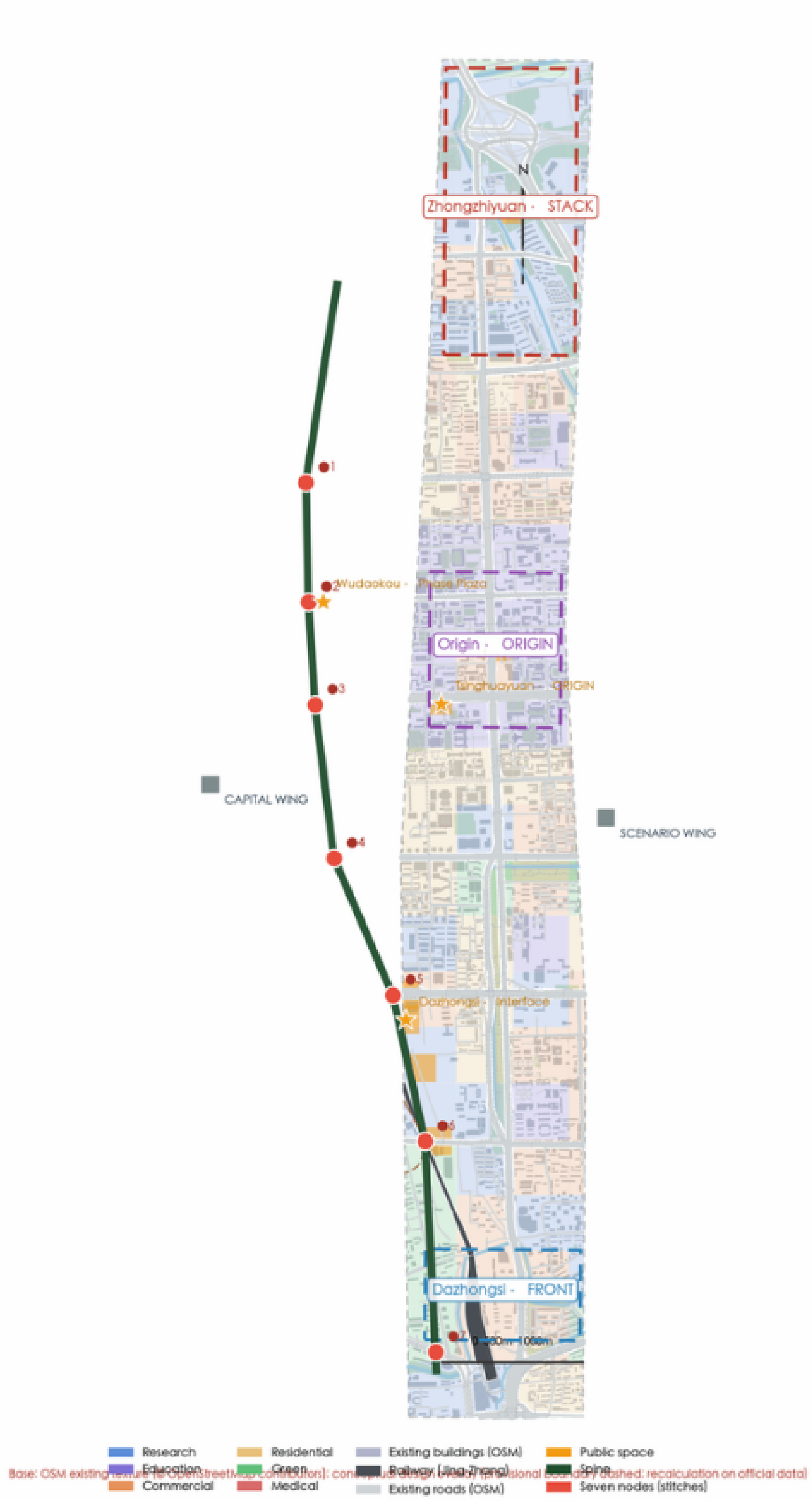
02

Spatial proposal: one line, three folds, two wings, seven nodes

Master plan · Key-area plans & sections · Character & height · Dazhongsi quadrants

⑤ Master plan (real fabric + concept overlay)

Fig.1 · Master Plan: one line, three folds, two wings, seven nodes (concept)

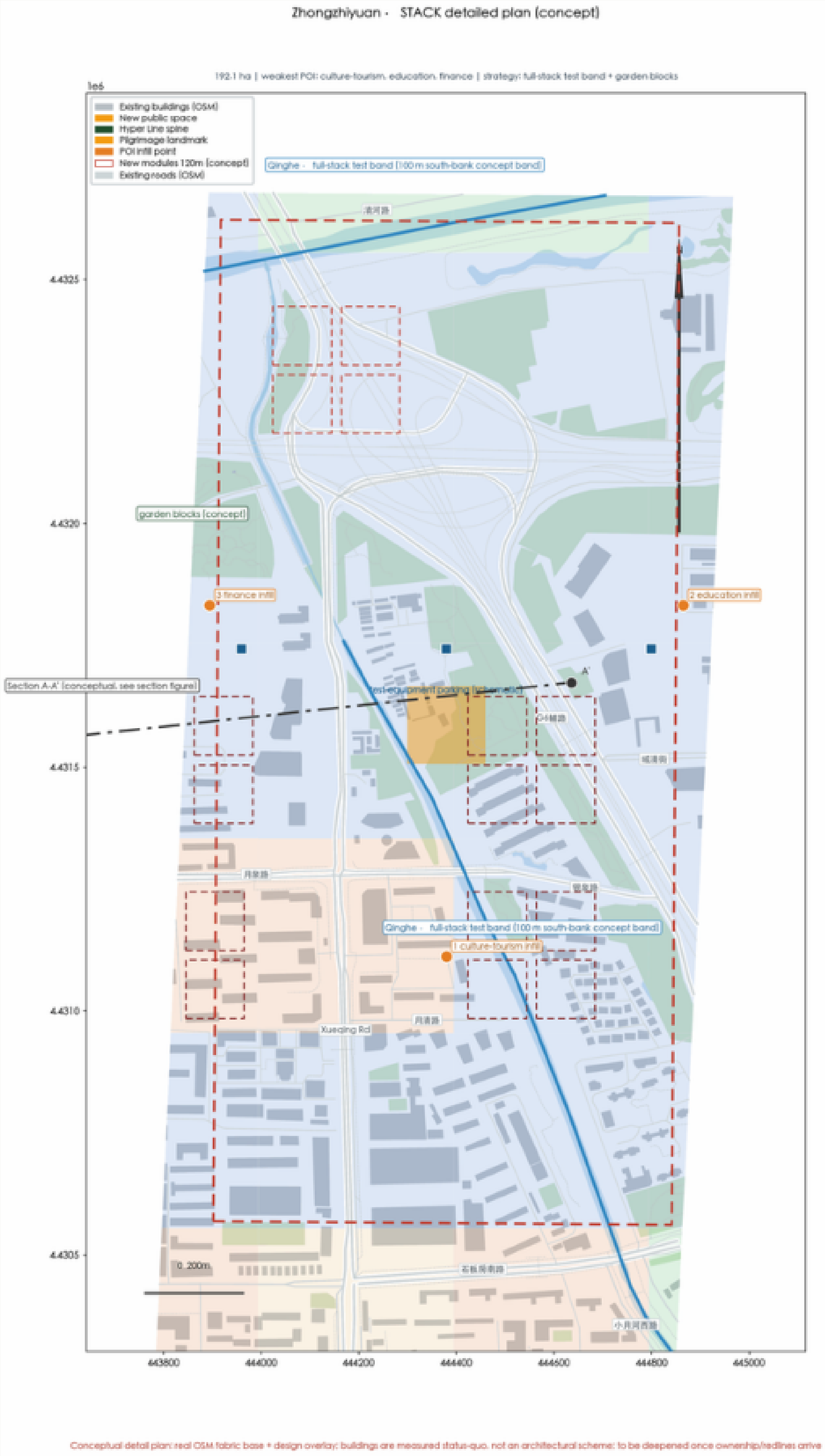


- Light grey = OSM building fabric; white-cased dark = real Jing-Zhang railway
- Dark green = Hyper Line spine; red dots = seven stitches; stars = three landmarks
- Three coloured frames = three gates: STACK / ORIGIN / FRONT
- The base is the real city; only the overlay is design

Base: OSM (© OpenStreetMap contributors)

7	3	2	11.4 km²
stitch nodes	key areas	wings	design area

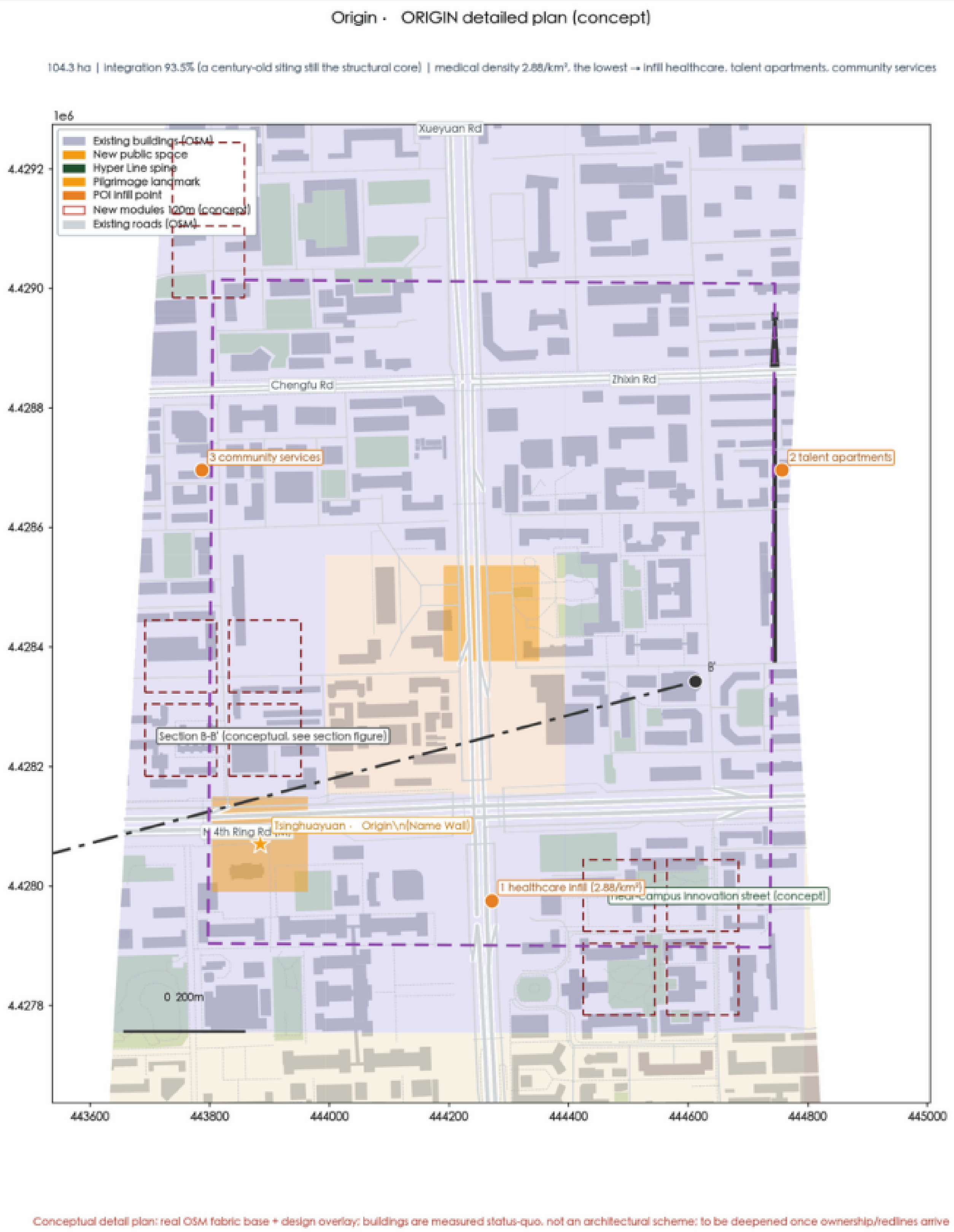
⑥ Zhongzhiyuan ·　STACK (192.1 ha)



- Qinghe full-stack test band: bounded tests in garden blocks
- Weakest: culture-tourism/education/finance → reasons to stay
- Section A-A': 3.2m footpath + 2.8m cycle band + test-equipment slots
- Long term (5–10y): garden-district renewal

192.1 ha	0.747	3
area (official)	jobs/life	infill categories

⑦ Origin · ORIGIN (104.3 ha)



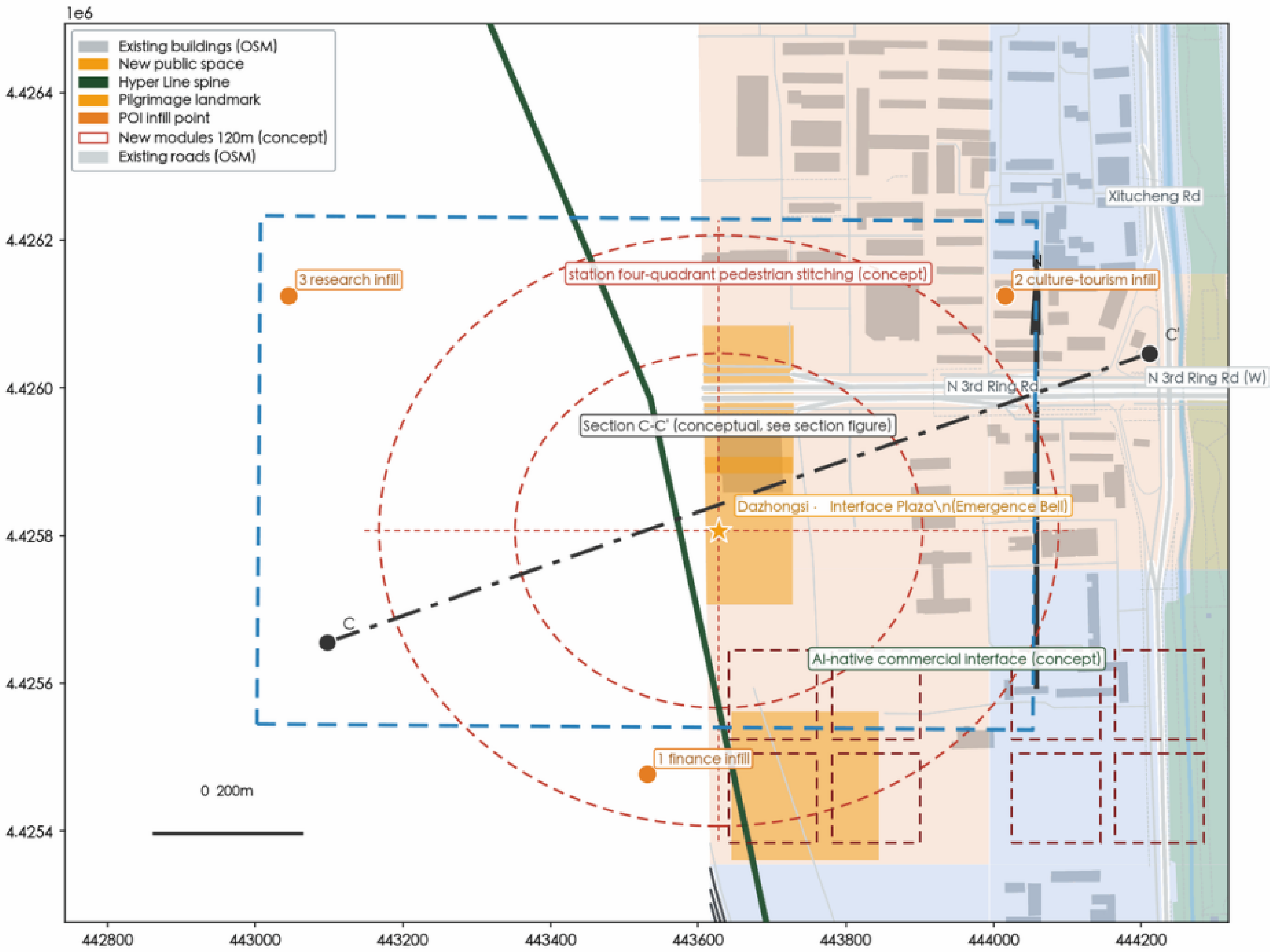
- Tsinghuayuan · Origin Plaza + Name Wall: integration 93.5%
- Wudaokou · Phase Plaza: the λ gauge — the city's thermometer
- Medical 2.88/km², the lowest → infill healthcare/apartments/services
- Mid term (3–5y): near-campus innovation street

104.3 ha	93.5%	2.88/km²
area (official)	structural origin	medical density

⑧ Dazhongsi · FRONT (72.0 ha)

Dazhongsi · FRONT detailed plan (concept)

72.0 ha | syntactic percentile 20.2% (southern periphery) | POI 511/km², the densest of the three · jobs-residence 0.344 | infill finance, culture-tourism, research



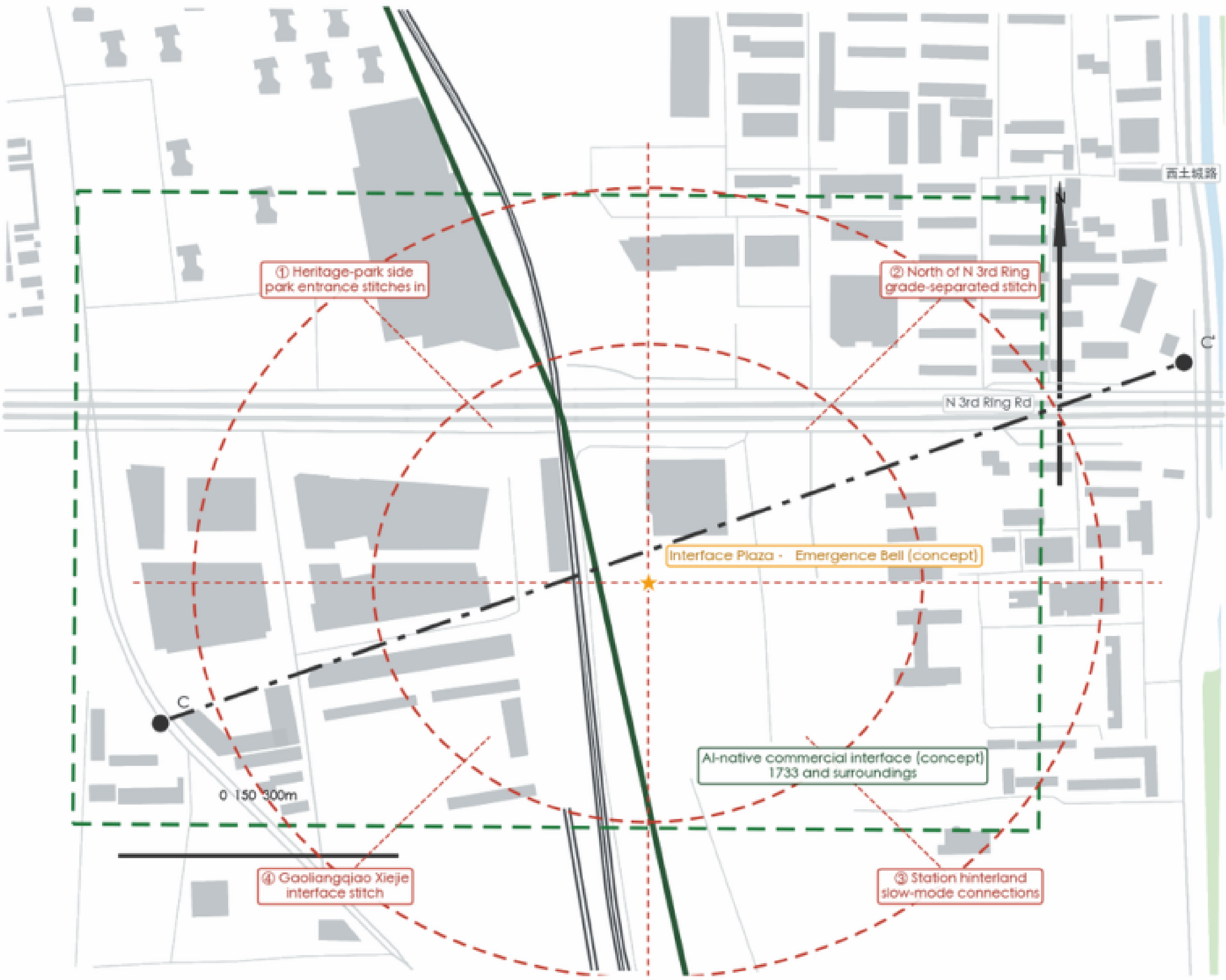
Conceptual detail plan: real OSM fabric base + design overlay; buildings are measured status-quo, not an architectural scheme; to be deepened once ownership/redlines arrive

- Syntactic periphery 20.2%, yet the densest POIs at 511/km²
- Jobs/life 0.344: life-service-led → infill finance/culture-tourism/research
- Light intervention, heavy interface: stitch four quadrants; nights for classes & merchants
- Near term (1–3y): spine stitching + Dazhongsi interface

72.0 ha	511/km²	0.344

⑨ Dazhongsi four-quadrant stitching (enlarged)

Fig C19 · Dazhongsi station-district four-quadrant stitching, enlarged (conceptual design)



Conceptual schematic: the stitch modes (grade-separation / slow-mode links) are conceptual suggestions, to be deepened once transport, municipal, and ownership data arrive; section C-C' shown in the detailed section figure.

Base: OSM (© OpenStreetMap contributors, ODbL)

- ① park entrance stitches in ② grade-separated stitch north of N3rd Ring
- ③ station-hinterland slow modes ④ Gaoliangqiao Xiejie interface
- Interface Plaza · Emergence Bell + the 1733 AI-native interface
- Section line C-C' matches the detailed plan

4 quadrant stitches | C-C' section line | 20m interface plaza

⑩ Street sections of the three areas (conceptual · dimension chain · plan index)

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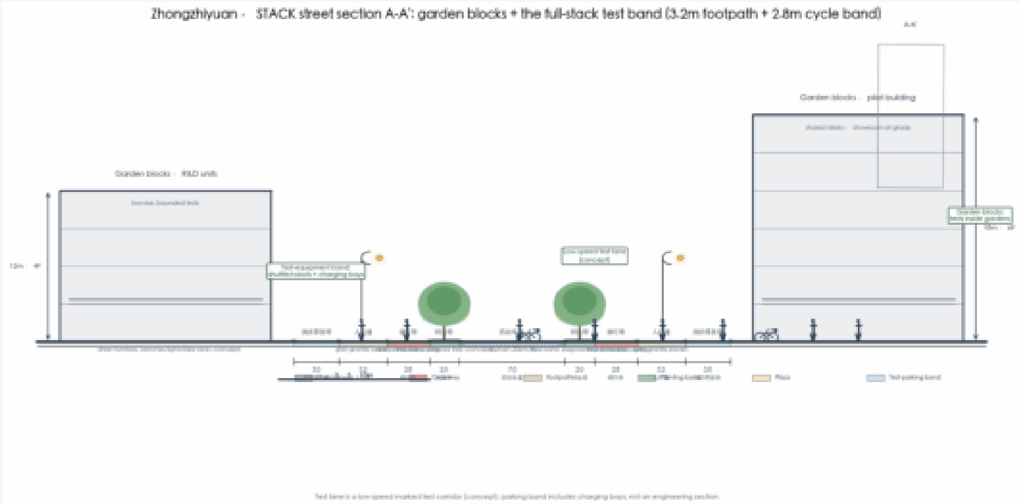
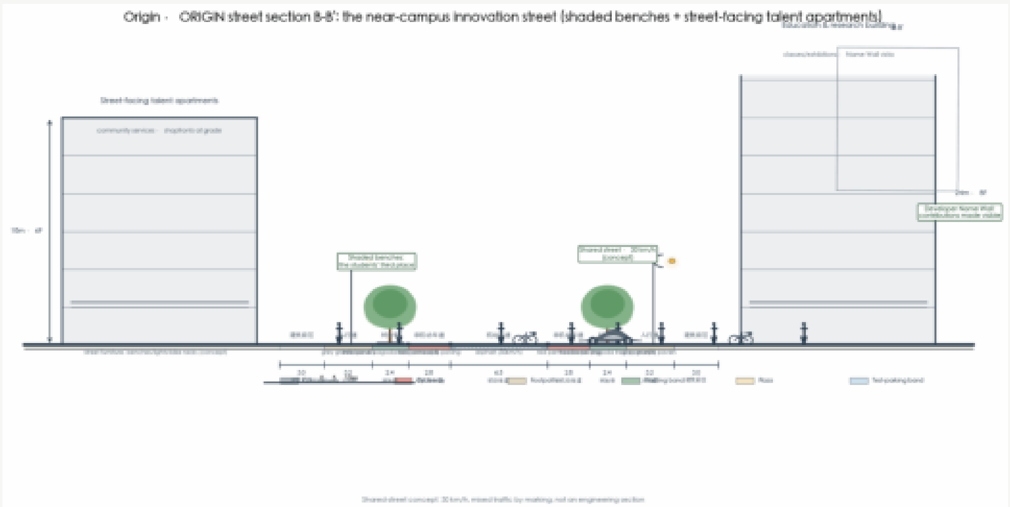
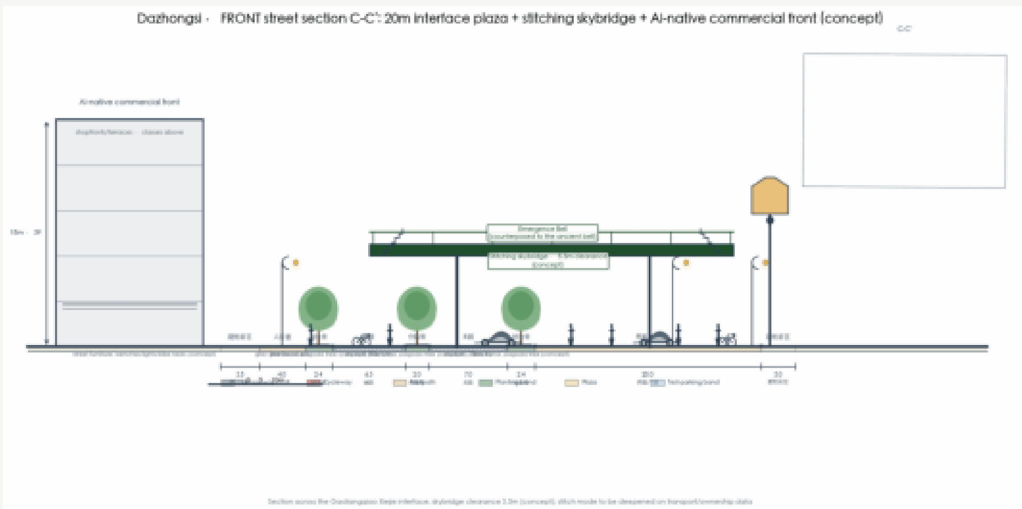
- Each section: building-to-building full section + dimension chain + floors/heights
- Scale references: 1.8m people · bikes · cars · low-speed shuttle · trees & lamps
- Dazhongsi C-C': 20m interface plaza + stitching skybridge (5.5m) + Emergence Bell
- Conceptual sections (not engineering); dimensions pending redline & heritage data







A-A' Zhongzhiyuan | B-B' Origin | C-C' Dazhongsi

Conceptual sections: design intent; to be deepened on redline, heritage and vertical data

Character & conceptual height zoning

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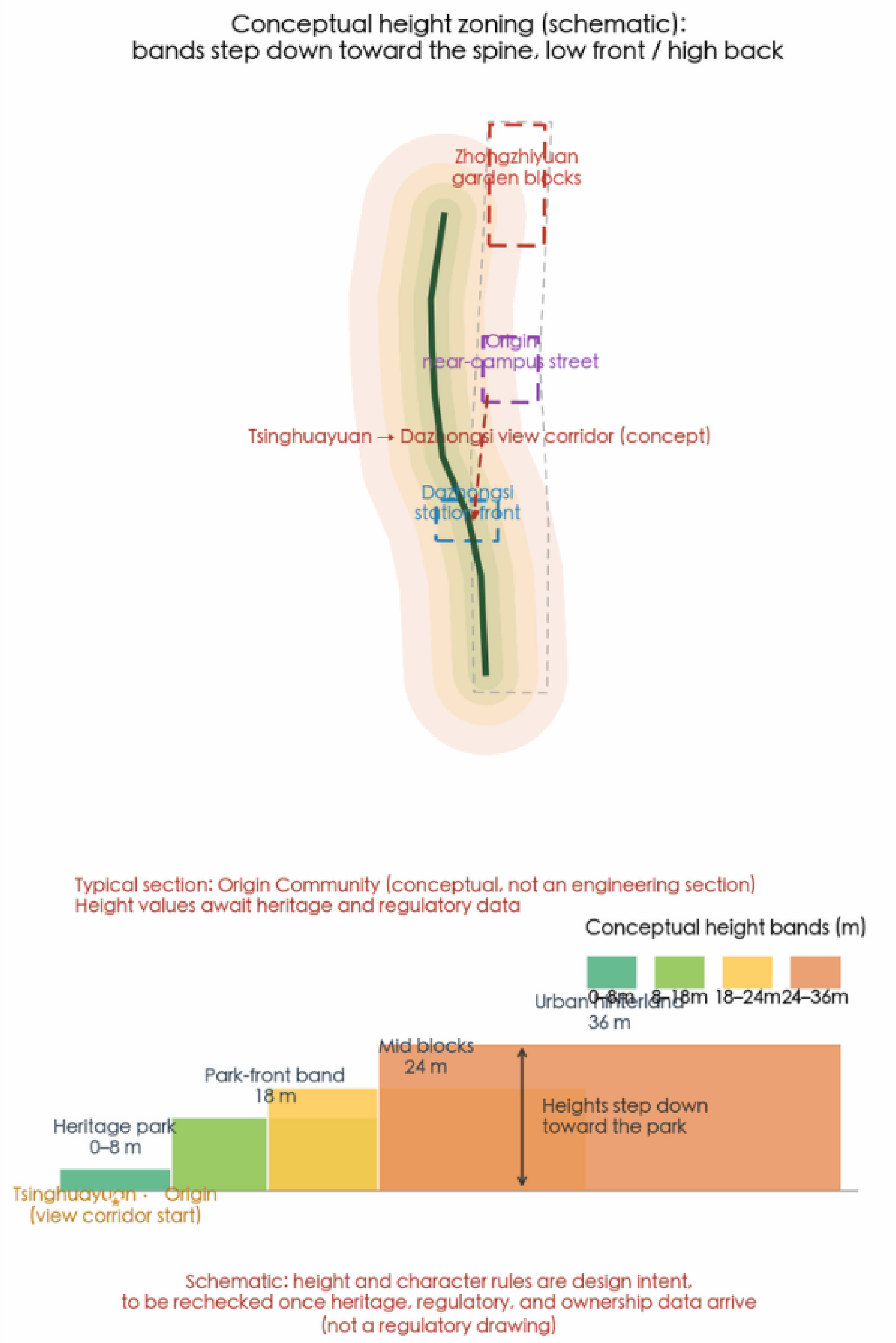


- Height bands step down along the spine: 0–8 / 18 / 24 / 36 m
- Three rules: garden blocks / near-campus street / station front
- Protective overlay (heritage + view corridors) + generative rules (low front, high back)
- Height values await heritage & regulatory data (not a regulatory drawing)

36m hinterland band	18m park-front	2 view corridors

03

Scenarios & ecosystem: the AI ecology and pilgrimage

Ecosystem map & seven cases · Ten scenario cards, five personas · Three pilgrimage landmarks · Operations & timeline

▯ AI innovation ecosystem map

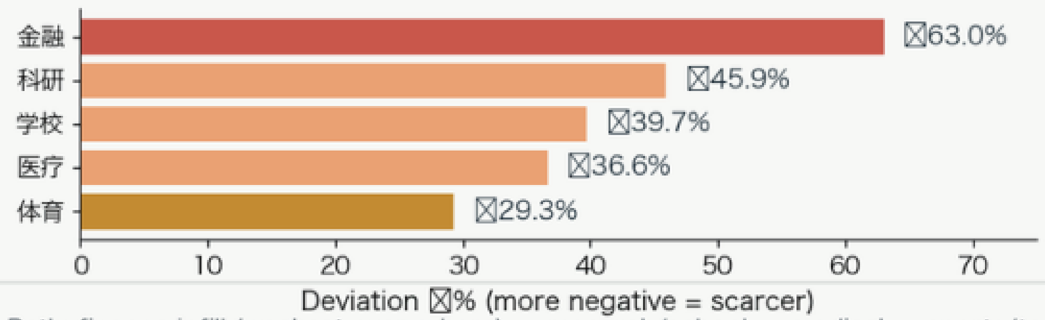
Fig · Industry ecosystem map: the 1+X+1 structure and five function gaps

Fill what is missing — finance is the weakest everywhere; the first ecosystem target (c8/c34 measured)

Structure 1+X+1

- 1 · flagship platform
- Zhongzhiyuan · full-stack
- Origin · origination

Five function gaps (c8 measured, ☒% vs uniform)



Path: finance infill (weakest everywhere) → research/school → medical → sports/tourism

Ecosystem maturity E_ecosys

design 0.85

Basis: deviation = (actual/uniform ☒1); E_ecosys = gap closure over five deficit categories (target ☒20%, provisional); diagnostics only, attraction is conceptual.

- Upstream complete: 1,000+ scientists, 13k developers, 180 OPCs, 32 chips
- Middle layer unmeasured: affordability/conversion/compute → annualized
- Seven cases, one transferable mechanism each (King's Cross / Kendall / Cambridge…)
- Five-party six-force conversion + the annual Phase-Forum report as interface

7 benchmark cases | 10 scenario cards | 5 personas

Fig · Operations timeline: stitch first, then infill and unlock; three phases with annual gates

A 90-day pilot first; each phase ends with the annual check-up (λ ⩾40% · H⩾1.35); no pass, no next phase

▼

Y1 - 3 · Stitching

Seven stitch connections + Dazhongsi front renewal; the 90-day pilot first (fence→paths→meter→operations)

▼

Y3 - 5 · Infill

Origin renewal: talent apartments delivered + finance/medical/research infill; live-work pairs (1:1)

Y5 - 10 · Unlock

Zhongzhiyuan garden blocks & full-stack test band: 42 CA cells re-allocated; Emergent Brain annualized

Phase gates: percolation 0.904 ☒ · syntax +36% ☒ · entropy +5.3% ☒ — or stop & review; money follows verification

Basis: phasing is conceptual (re-sequenced on official data); gates share sources with the three gates (c2/c4b/c28 measured).

- Annual: dev festival / Phase Forum (λ & H report) / open test seasons
- Attraction funnel: OPC contest → order plaza → desks → incubator → conversion
- Two acceptance gates: λ⩾40%, H⩾1.35 — or stop and review
- Phases: 1–3 stitching+interface / 3–5 Origin / 5–10 Zhongzhiyuan

2	5	10年
acceptance gates	annual events	timeline

International benchmark matrix: cases × mechanisms × responses × numbers

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- Seven global cases, one transferable mechanism each (no copied images or geometry)
- King's Cross × stitch-first / Kendall × conversion circle / Stanford × garden neighbourhoods
- Yuehai × dense fabric / Shibuya × consumption / Cambridge × symbiosis / Rothschild × public life
- The opportunity: seven mechanisms hold at once on one belt — Jing-Zhang's unique window

Fig C26 · International benchmark matrix: seven global cases, one transferable mechanism, one of our measured numbers

Benchmark discipline: no copied images or geometry — mechanisms only; each mechanism lands on one of our numbers

Case	Mechanism	Our response	Measured
London · King's Cross	open the paths and squares first, let the plots grow themselves	stitch the seven nodes first, then the plots	integration +43%–+70%
Boston · Kendall Square	the near-campus conversion circle: density within 1 km of MIT	Origin near-campus street: catching those leaving school	Tsinghuayuan integration 93.5%
Stanford Research Park	campus origination: low-density garden research neighbourhoods	Zhongzhiyuan garden blocks + the full-stack test band	13,000 developers · 32 chips
Shenzhen · Yuehai Street	the fabric density of high-density startup blocks	Dazhongsi front: a 511 POI/km² station interface	25,476 POIs measured
Tokyo · Shibuya	the renewal paradigm of a content-consumption interface	AI-native commercial front + Emergence Bell (night classes/exhibitions)	10 scenario cards landed
Cambridge Cluster	the symbiotic structure of a university town and an industry belt	the five-party six-force conversion + the annual Phase-Forum report	180 OPCs · 18 vendors adapted
Tel Aviv · Rothschild Blvd	the public life of a startup boulevard	the dev festival + honor wall + spine public space	equity loops 5/5 online

The opportunity: seven mechanisms all hold on one belt — the unique opportunity of Centennial Jing-Zhang: the world's first AI belt designed by urban science and run recursively.

Benchmark discipline: mechanisms transferred, numbers computed locally

7 benchmark cases

7 transferable mechanisms

H=1.556 investability

04

System & finance: the Emergent Brain and the investment ledger

Emergent Brain architecture · Phase Control Room · Investment & return · Risk & compliance

▯ The Emergent Brain: the city as a recursive program

- Three layers: sense → compute → decide, with humans always holding the veto
- Four terminals: citizens / enterprises / government / campus
- The recursive loop is the annual check-up: read→compute→tune→verify→re-read
- Red lines: no privacy violation, no over-monitoring, human-reviewable

3 layers	4 terminals	5 equity loops

Phase Control Room: the belt's annual check-up

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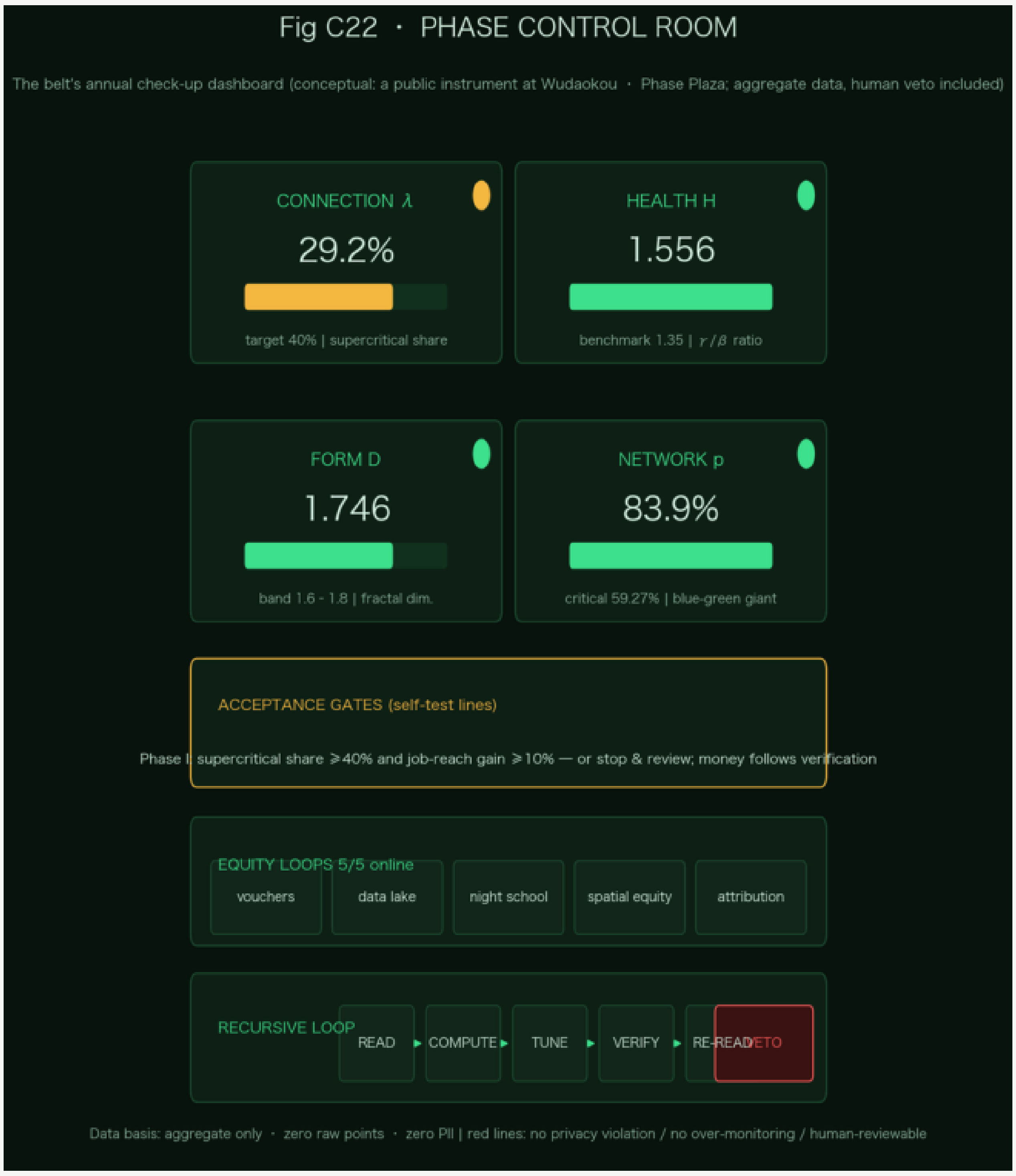
- Four gauges: λ 29.2%→40% target / H 1.556 / D 1.746 / p 83.9%
- Live panels: job reach 323→375, match 46.2%→61.5%, η +60%
- Physically at Wudaokou · Phase Plaza: a public instrument, not surveillance
- Aggregate only, zero raw points, zero PII, human veto

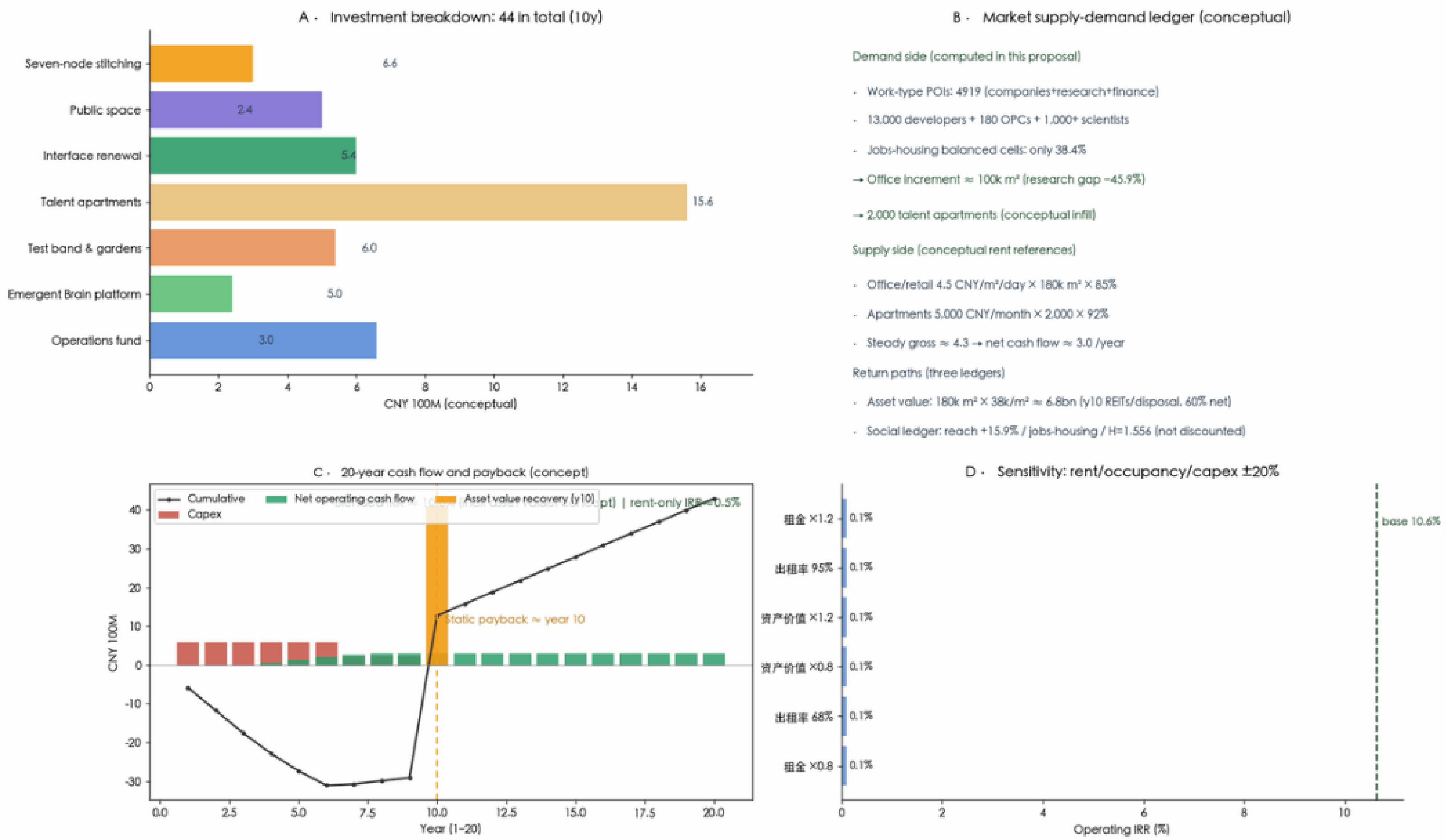


4 critical gauges	25,476 data base	5/5 equity online

▯ Investment & return (conceptual magnitude)

- CNY 4.40bn: stitching 0.66 / space 0.24 / interface 0.54 / apartments 1.56 / test band 0.60 / brain 0.50 / ops 0.30

Fig C20 · Investment budget and return estimation (conceptual magnitude, not an investment commitment)



Basis: conceptual estimation (public unit-price magnitudes + demand computed in this proposal): rents and occupancy are market-reference assumptions, not a return commitment; precise estimation follows the official redline, regulatory plan, and status-quo data.

- Three ledgers: rent (IRR≈0.5% alone) + asset (y10 REITs/disposal, blended IRR≈10.6%) + social
- Demand: 4,919 work POIs → 100k m² offices; 38.4% balance → 2,000 apartments
- Public ledger ≈1.7bn seeks no financial return; market ledger ≈2.7bn runs rent+asset

Conceptual; not an investment commitment; rents & occupancy are replaceable assumptions

44.0亿 total (concept)	10.6% blended IRR	10年 static payback

05

Conclusion: four sentences, each with numbers

Conclusion dashboard · 1909 switch → 2026 belt

Design-conclusion dashboard

- ① Re-grow, not rebuild: D=1.746 / H=1.556 / giant 83.9%
- ② Move aside what blocks: 7 gaps→7 stitches / 42 cells / three annualized
- ③ Falsifiable: jobs +15.9% / match +15.3pct / η +60%
- ④ An OS, not a blueprint: four rulers + brain + check-up + exit

4 conclusions	2 gates	H≥1.35 health line

Move aside what blocks emergence, and leave the rest for the city to grow itself.

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KUN-SAL Spatial Quant Lab · Xing Xinhua · GitHub: Anshengdesign